Tested Prompt and Output

AI Quality Incident Assistant for Hertzmann & Söhne

Course: AI Orientation & Discovery – Final Project

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Purpose of this document

This document provides evidence supporting the proposed **AI Quality Incident Assistant** described in the AI Strategy Proposal.

The purpose of the testing is to evaluate whether a Large Language Model (Claude) can reliably convert unstructured quality incident notes into a standardized first draft of a Non-Conformance Report (NCR).

The AI assistant is intended to support quality engineers by:

- extracting relevant factual information,
- organizing the information into a standardized report,
- identifying missing information,
- assigning a preliminary incident category.

The assistant **does not replace the quality engineer** and **does not make compliance decisions**. Every generated report requires review and approval by a qualified engineer before use.

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1. AI Assistant Workflow

The proposed workflow is illustrated below.

Quality Engineer

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Raw Incident Notes

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Claude AI

• Extract facts

• Classify incident

• Detect missing information

• Draft NCR

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▼

Quality Engineer Review

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Final Approved NCR

The workflow ensures that AI is used only for documentation support while maintaining human oversight for all technical and regulatory decisions.

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2. Prompt Design

The prompt was designed following prompt engineering best practices introduced during the course.

It contains five clearly separated sections:

- Role
- Context
- Instructions
- Constraints
- Output Format

Separating these components helps reduce ambiguity and improves output consistency.

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3. Final Prompt Used for Testing

You are a quality-documentation assistant supporting Hertzmann & Söhne, a German precision-parts manufacturer.

<context>

Hertzmann prepares non-conformance reports as part of its ISO 9001 and IATF 16949 quality processes. Engineers currently prepare these reports manually from inspection and incident notes.

Your output is only a first draft. A qualified quality engineer will verify all facts, confirm or correct the incident category, complete missing information, and approve the final report.

</context>

<source_notes>

{{Paste the engineer's raw incident notes here}}

</source_notes>

<instructions>

Read the source notes and perform the following tasks:

1. Extract only facts explicitly stated in the notes.

2. Assign one preliminary incident category from this list:

- Dimensional deviation

- Material issue

- Surface defect

- Process deviation

- Documentation issue

- Packaging issue

- Other

3. Identify information that is missing from the notes.

4. Create a structured first draft of the quality incident report.

Extract the following fields:

- Part or product identifier

- Batch or lot number

- Detection stage

- Incident category

- Non-conformance description

- Required specification or tolerance

- Actual measured value

- Immediate containment action

- Follow-up explicitly mentioned

- Missing information

Do not invent missing facts.

Do not determine the root cause.

Do not assign final severity.

Do not decide whether the product is compliant.

Do not create corrective actions that are not stated in the notes.

Do not change numerical values, units, part numbers, or batch numbers.

If a value is unavailable, write "Not provided."

</instructions>

<output>

Return one plain-text table with exactly two columns:

Field | Extracted Information

Use exactly these rows:

Part or Product Identifier

Batch or Lot Number

Detection Stage

Preliminary Incident Category

Non-Conformance Description

Required Specification or Tolerance

Actual Measured Value

Immediate Containment Action

Follow-up Mentioned in Notes

Missing Information

After the table, add one line only:

Engineer Review Required: Yes

Do not include an introduction.

Do not include recommendations.

Do not include conclusions.

Do not include commentary outside the table and review line.

</output>

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4. Test Methodology

To evaluate robustness, the prompt was tested using **three different fictional quality incidents** representing common manufacturing situations.

The objective was to determine whether the assistant could consistently:

- preserve technical measurements,
- preserve identifiers,
- correctly classify incidents,
- identify missing information,

- avoid hallucinations,
- follow the requested report structure.

All test cases use fictional data created solely for this academic project.

Test Case 1 – Dimensional Deviation

Objective

Verify that the assistant correctly extracts dimensional measurements while preserving all numerical values and tolerances.

Input

Engineer Notes

Part 744A failed dimensional inspection during final inspection. Measured diameter: 10.42 mm. Maximum permitted diameter: 10.30 mm. Affected batch: 24-071. Production was paused and the batch isolated. Supplier material certificate requires review. Root cause has not yet been identified. Corrective-action owner and completion deadline have not yet been assigned.

AI Output

Field	Extracted Information
Part or Product Identifier	Part 744A
Batch or Lot Number	24-071
Detection Stage	Final inspection
Preliminary Incident Category	Dimensional deviation
Non-Conformance Description	Part 744A failed dimensional inspection during final inspection; measured diameter exceeded the maximum permitted diameter.
Required Specification or Tolerance	Maximum permitted diameter: 10.30 mm
Actual Measured Value	10.42 mm
Immediate Containment Action	Production was paused and the batch isolated.
Follow-up Mentioned in Notes	Supplier material certificate requires review.
Missing Information	Root cause; corrective-action owner; completion deadline.

Engineer Review Required: Yes

Screenshot

(Insert Screenshot 2 – Claude response)

Evaluation

Criterion	Result
Part number preserved	✓ Pass
Batch preserved	✓ Pass
Measurements preserved	✓ Pass
Category appropriate	✓ Pass
Missing fields identified	✓ Pass

Criterion	Result
Invented information	None
Output format followed	✓ Pass

Discussion

The assistant successfully extracted all numerical values without modification and correctly identified the incident as a dimensional deviation. Missing information was explicitly listed rather than invented, demonstrating compliance with the prompt constraints. The structured output required only human verification before inclusion in the final NCR.

Test Case 2 – Surface Defect

Objective

Verify that the assistant can classify cosmetic defects while avoiding unsupported conclusions.

Input

Incoming inspection identified scratches on the machined surface of part 883C.

Four components exceeded the customer's cosmetic specification.

Batch 25-014 was quarantined.

Supplier Quality has been informed.

Root cause investigation is pending.

No corrective actions have been defined.

Actual Claude outputs

Screenshot

Field	Extracted Information
Part or Product Identifier	Part 883C
Batch or Lot Number	Batch 25-014
Detection Stage	Incoming inspection
Preliminary Incident Category	Surface defect
Non-Conformance Description	Scratches identified on the machined surface of part 883C; four components exceeded the customer's cosmetic specification.
Required Specification or Tolerance	Not provided
Actual Measured Value	Not provided
Immediate Containment Action	Batch 25-014 was quarantined.
Follow-up Mentioned in Notes	Supplier Quality has been informed.
Missing Information	Required cosmetic specification/tolerance; actual measured/observed values for the four components; root cause; corrective actions.

Engineer Review Required: Yes

Evaluation

Criterion	Result
Part number preserved	✓ Pass
Batch preserved	✓ Pass
Surface defect recognized	✓ Pass
Missing information detected	✓ Pass
Hallucinations	None
Formatting followed	✓ Pass

Discussion

The assistant correctly differentiated a cosmetic defect from a dimensional deviation. It avoided assigning a root cause or suggesting corrective actions that were not mentioned in the source notes.

Test Case 3 – Documentation Issue

Objective

Verify that the assistant correctly recognizes documentation-related non-conformities.

Input

Incoming inspection revealed that the supplier material certificate was missing.

Batch RM-2026-118 was placed on hold.

No dimensional or surface defects were observed.

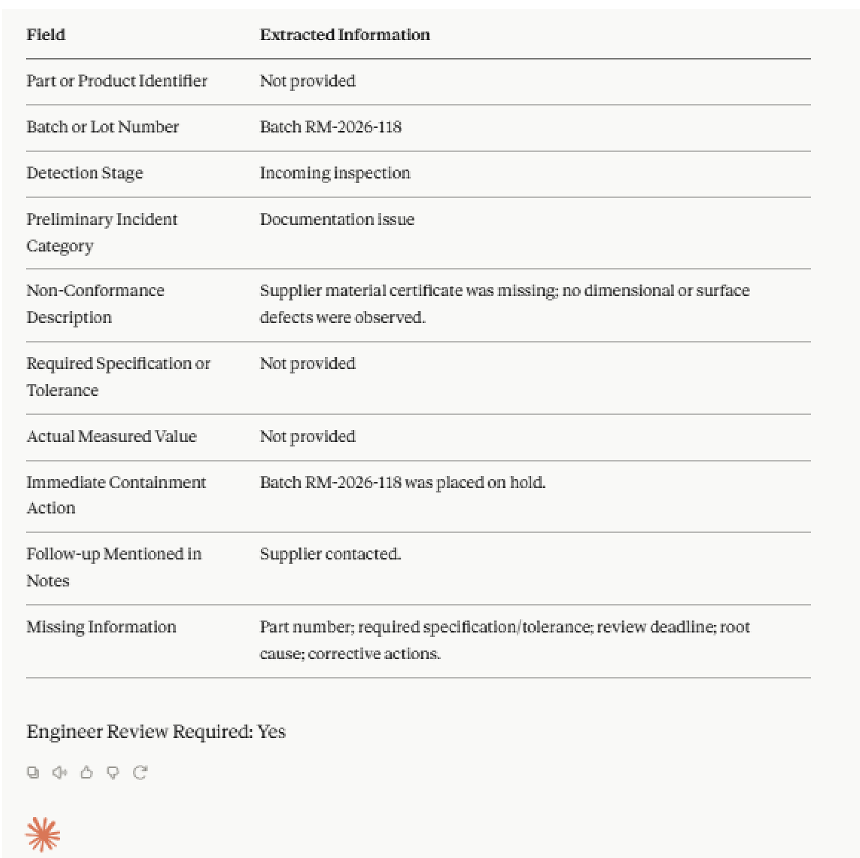
Supplier contacted.

Review deadline not assigned.

Part number not recorded in the notes.

Actual Claude outputs

Screenshot



Field	Extracted Information
Part or Product Identifier	Not provided
Batch or Lot Number	Batch RM-2026-118
Detection Stage	Incoming inspection
Preliminary Incident Category	Documentation issue
Non-Conformance Description	Supplier material certificate was missing; no dimensional or surface defects were observed.
Required Specification or Tolerance	Not provided
Actual Measured Value	Not provided
Immediate Containment Action	Batch RM-2026-118 was placed on hold.
Follow-up Mentioned in Notes	Supplier contacted.
Missing Information	Part number; required specification/tolerance; review deadline; root cause; corrective actions.

Engineer Review Required: Yes

Evaluation

Criterion	Result
Documentation issue identified	✓ Pass
Missing part number detected	✓ Pass
Missing review deadline detected	✓ Pass
No invented facts	✓ Pass
Formatting correct	✓ Pass

Discussion

The assistant correctly classified the incident as a documentation issue despite the absence of a physical product defect. It identified missing fields without attempting to infer unavailable information.

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5. Prompt Refinement

The prompt was tested iteratively to improve output consistency.

Version 1

Observed Issues

During the first test, the model occasionally:

- added introductory text before the table,
- produced recommendations outside the requested format,
- returned inconsistent field names.

Although the extracted information remained accurate, these inconsistencies reduced the suitability of the output for standardized documentation.

Improvements Introduced

The following constraints were added:

- Return only the requested table.
- Do not include commentary.
- Do not generate recommendations.
- Do not determine root cause.
- Do not decide product compliance.
- Preserve all numerical values exactly.
- If information is unavailable, write "Not provided."

Version 2

After refining the prompt, the outputs became significantly more consistent.

The assistant:

- consistently returned the requested table,
- preserved all measurements,
- avoided hallucinations,
- listed missing information correctly,
- followed the requested formatting.

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6. Overall Validation Results

Test Case	Classification	Result
Test Case 1	Dimensional deviation	✓ Pass
Test Case 2	Surface defect	✓ Pass

Test Case	Classification	Result
Test Case 3	Documentation issue	✓ Pass

Overall Performance

Criterion	Result
Measurements preserved	100%
Technical identifiers preserved	100%
Incident classification appropriate	100% (three representative test cases)
Missing information identified	100%
Invented facts	None observed
Required output structure followed	100%

These percentages reflect the outcomes of the three fictional test cases used in this academic project. They are not intended to represent production-scale performance.

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7. Limitations

Although the prompt performed well on the selected test cases, several limitations remain.

The AI assistant:

- cannot verify the technical correctness of engineer notes,
- may misclassify complex or ambiguous incidents,
- cannot determine regulatory compliance,
- cannot perform root-cause analysis,

- should not generate corrective actions,
- must always operate under human supervision.

Additional testing using anonymized historical NCRs would be required before deployment in a production environment.

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8. Conclusion

The prompt testing demonstrates that the proposed AI Quality Incident Assistant can consistently transform unstructured quality incident notes into standardized draft reports while preserving critical technical information and identifying missing data.

The assistant showed consistent performance across three representative quality scenarios, including dimensional deviations, surface defects, and documentation issues. Throughout testing, it complied with the defined constraints by avoiding unsupported conclusions, preserving numerical values, and requiring human review for every report.

These results support the feasibility of using generative AI as a documentation assistant within Hertzmann & Söhne's quality management process. However, the solution is intended only as a decision-support tool. Final responsibility for technical interpretation, regulatory compliance, and report approval remains with qualified quality engineers.